# stolperbeats

## _travelers guide
## _for stumbling across the universe

**6** tracks + subdiv | **24** patterns per bank | **7** shuffles × 4 × 4

› then  noon = 12 o'clock  tags = the panel drawing

shuffled drum trigger sequencer · six tracks + subdiv clock · 7 shuffles × 4 timings × 4 feelings · firmware 162 manual

## 01 what it is

sb_01

| | |
|---|---|
| a drum sequencer | trigger patterns for `kick` `snare` `hihat 1` `hihat 2` `perc 1` `perc 2` + a `subdiv` clock |
| the groove | 7 shuffles × 4 timings × 4 feelings, laid over the same patterns |
| the patterns | 24 per bank per track · 64 banks, 4 active out of the box |

## 02 press play

sb_02 · tempo encoder

| | |
|---|---|
| `[ tempo ]` click | start / stop |
| `[ tempo ]` turn | 15 – 250 bpm in steps of one · nudge to show the tempo |
| `[ tap ]` | tap quarter notes to set the tempo |
| `sync` in | a clock here wins over the encoder · the module latches onto its ticks |
| `reset` | a trigger sends the sequence back to step 1 |

## 10 first flight

sb_10 · first beat

1. `kick` `snare` `hihat 1` trigger outs into three drums
2. click `[ tempo ]` to start · turn it to taste
3. turn `[ kick ]` and `[ snare ]` through their 24 patterns
4. turn `[ shuffle ]` down the matrix
5. press `[ shake ]`, then press again for each feeling
6. try `[ trip ]`, `[ push ]` and `[ clave ]` on the same beat

## 03 patterns

sb_03 · instrument encoders · fig.02

| | |
|---|---|
| turn | pick one of the 24 patterns in the bank for that track |
| click | mute / unmute the track |
| kick · snare | 4 lines × 16 steps = 4 bars each |
| hihats · percs | 1 line × 16 steps = 1 bar each, one line per track |
| one step | a 16th note in 4/4, before shuffle, timing and feeling move it |

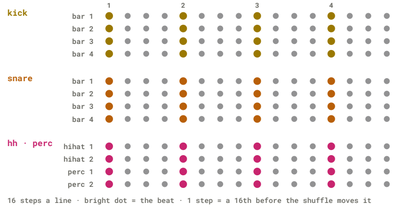
## 15 midi in

sb_15 · fixed channels

| | |
|---|---|
| ch 1 – 6 | kick, snare, hihat 1, hihat 2, perc 1, perc 2 · any note steps through the 24 patterns |
| ch 7 | subdiv: shuffled 16th ↔ n-tuplet |
| ch 8 | steps through the 7 shuffles |
| ch 9 | steps through the feelings · every 4 notes moves on a timing |

## __fig.02 : the pattern matrix

right matrix · 64 / 64 / 4 × 16




16 steps a line · bright dot = the beat · 1 step = a 16th before the shuffle moves it

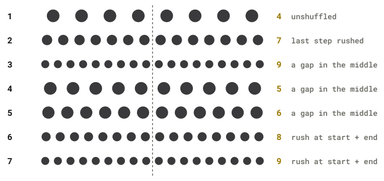
## 04 shuffle

sb_04 · shuffle encoder · fig.01

| | |
|---|---|
| turn | 7 shuffles · the further down the left matrix, the stronger |
| the display | how 8 steps (2 × 4) spread in time · the dots per line are subdivisions |
| the steps | always 8 per line · only the spacing changes |
| `shuffle` cv | pick the shuffle with cv |

## __fig.01 : the shuffle display

left matrix · 7 lines




dots = subdivisions per 4 steps (trip timing) · lower line = stronger shuffle

## 05 timing · feeling

sb_05 · trip · shake · push · clave

| | |
|---|---|
| `[ trip ]` | n-tuplet grooves tripping over themselves |
| `[ shake ]` | golden-age hip hop 8th-note swing |
| `[ push ]` | gently makes time for the 4/4 downbeats |
| `[ clave ]` | 3:3:2 stays fixed, the rest swings around it |
| press again | next feeling, shown by the backlight colour: `tight` everything shuffles together |
| rolls | hihat 2 becomes a trigger burst |
| flam | kick and snare land very slightly late |
| slop | flam, with the hihats following |

## 06 edit a pattern

sb_06 · instrument encoders

| | |
|---|---|
| enter | long press + release that track's encoder |
| turn · click | move the cursor · set or clear the step |
| leave | long press the encoder again, then let go … |
| › fast blink | … now: changes thrown away |
| › slow blink | … wait for it: changes saved |

## 07 outputs

sb_07 · top row · 10 V

| | |
|---|---|
| `kick` … `perc 2` | trigger out per track · into a drum's trigger in |
| `subdiv` | turn the encoder: a shuffled 16th clock or a regular n-tuplet of the shuffle · clock a 1-tick-per-step sequencer |
| `sync` | unswung clock · ppqn 1 – 10, 12, 16, 24 or once per 1, 2, 4 bars (resets, 4/4) |
| `env` `env` | two tempo-synced cv outs · section 11 |
| drum voice | play disting ex samples from these outs · `disting ex wormhole` |

## 08 clocking

sb_08 · sync encoder

| | |
|---|---|
| `[ sync ]` click | pick what to edit: in (top row, red) or out (middle row, orange) |
| sync in | 1, 2, 3, 4, 6, 8, 12, 16 or 24 ppqn |
| first / last pulse | an incoming clock starts the sequence, and stops it when it stops (config) |
| midi clock | 24 ppqn in and out · play sends start / stop |

## 11 envelopes

sb_11 · press + release subdiv 2 s

| | |
|---|---|
| `env 1` | `[ kick ]` wave · `[ snare ]` clock div · `[ hihat 1 ]` low · `[ hihat 2 ]` high |
| `env 2` | `[ perc 1 ]` wave · `[ perc 2 ]` clock div · `[ subdiv ]` low · `[ sync ]` high |
| 9 waves | ramp down · ramp up · tri up · tri down · sine · cosine · pulse high · pulse low · stepped |
| clock div | x4 · x2 · x1 bars · .5 bar · 1 beat · or bd sd h1 h2 p1 p2 = one cycle per hit |
| :bd … :p2 | per-hit velocity · high = accent level, low = non-accent level |
| leave | press + release subdiv 2 s · saved when you stop the sequence |

## 12 banks

sb_12 · ◄ bank ►

| | |
|---|---|
| `[ ◄ ]` `[ ► ]` | step through the active banks · 64 banks of 24 patterns |
| bank edit | press + release a bank button 2 s · same again to leave |
| `[ kick ]` turn · click | pick a bank (blinking) · set it active or inactive |
| inactive | skipped by the bank buttons · keeps its patterns |
| why | perform on a different set · A / B / fill variations · blank slates |
| `bank` cv | pick the active bank with cv |

## 13 cv · linear

sb_13 · second row

| | |
|---|---|
| `pattern cv` | one per track · picks a pattern from the active bank |
| `shuffle` `bank` | cv picks the shuffle and the bank · all 0 – 5 V (jumpers: 10 V) |
| `[ linear ]` | thins out overlapping steps so one hit plays at a time |
| who wins | snare › kick › hihat 2 › hihat 1 · snare › kick › perc 2 › perc 1 |

## 14 midi out

sb_14 · trs · per-track notes

| | |
|---|---|
| `midi out` | trs type a · rear switch for b · in takes a or b |
| 7 tracks | channel, note, velocity, duration each |
| midi edit | hold `[ shuffle ]` 2 s, let go · slow blink |
| midi rec | hold 3 s · fast blink · incoming midi sets note + channel |
| show · set | nudge a track encoder · click picks, turn changes |
| `[ ◄ ]` `[ ► ]` | 32 midi banks (drum machine maps) · hold `[ ◄ ]` = save |

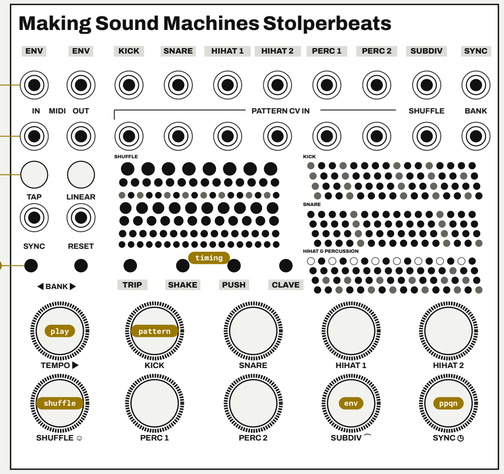
## 16 card · expander

sb_16

| | |
|---|---|
| sd card | patterns, settings and midi banks · written when you stop · no card = defaults every power-up |
| MachineConfig | play-position chaser · clock starts / stops · reset toggles play · midi in, clock and transport on / off |
| expander | 3 hp, passive · linear + mute per track and subdiv · try euclidean or random gates |

## __fig.03 : the panel

numbers match the legend




`play` `tempo` · click = start / stop · turn = bpm
`pattern` `kick … perc 2` · turn = pattern · click = mute · hold = edit
`shuffle` `shuffle` · 7 shuffles · hold 2 / 3 s = midi edit / rec
`env` `subdiv` · shuffled 16ths or n-tuplet · hold 2 s = envs
`ppqn` `sync` · clock divider · click = in / out
`timing` `trip … clave` · timing · press again = next feeling
`banks` `◄ bank ►` · active banks · hold 2 s = bank edit
`tap` `tap · linear` · tap tempo · one hit at a time
`outs` `top row` · env · triggers · subdiv · sync out
`midi` `second row` · midi in / out · pattern, shuffle, bank cv

## 09 specs

sb_09

| | |
|---|---|
| size | 26 hp + 3 hp passive expander · 35 mm deep |
| power | 280 mA +12 V · 13 mA –12 V · expander: no power |
| outs | triggers, clocks and envs at 10 V |
| cv in | 0 – 5 V · rear jumpers for 0 – 10 V |
| storage | class 10 fat32 sd card · firmware over usb |